Day 10 — TNPSC Group 2A Study Material

Part A – General Studies: Geography of India: Monsoon, Rainfall, Weather & Climate

READING MATERIAL

Second sub-topic of Geography Unit II. Real exam evidence favors specific named facts (instrument names, place names, named cyclones) over general monsoon theory — know the specifics.

Q: Real exam (Assertion-Reason): "The eastern coast of India is affected by tropical cyclones more than the western coast." Is the reason "tropical cyclones form only in the Bay of Bengal" correct?

A: The Assertion is TRUE (the eastern/Bay-of-Bengal-facing coast is indeed hit harder), but the Reason is FALSE — cyclones form in both the Bay of Bengal AND the Arabian Sea, just more frequently and intensely in the Bay of Bengal (due to warmer sea surface temperatures and wind patterns there).

Q: A sudden fall in barometer reading indicates what kind of weather is coming?

A: A storm.

Q: Which instrument measures humidity?

A: A Psychrometer (not a hydrometer, altimeter, or radiometer — those measure liquid density, altitude, and radiation respectively).

Q: What causes North India's continental climate (large temperature swings between seasons)?

A: Its location far away from the moderating influence of the seas — not sun-ray angle or simply being on plains.

Q: Where in India do you find a Meso-thermal climate?

A: The Gangetic plain.

Q: Which Indian states are most affected by tropical cyclones?

A: Odisha and Andhra Pradesh (on the Bay of Bengal coast).

Q: Where is the highest temperature in India typically recorded?

A: Ganga Nagar (Rajasthan).

Q: Which cyclone is nicknamed the 'Quirky Cyclone'?

A: Cyclone Gaja.

Q: Where does the Chinook wind blow?

A: North America.

Q: At what wind speed does the World Meteorological Organisation officially recognize a storm as a cyclone?

A: Exceeding 63 kph.

Q: What are 'Twisters' otherwise known as, and what are winds that blow from one fixed direction all year called?

A: Twisters = Tornadoes. Winds blowing consistently from one direction year-round are called Planetary winds (distinct from seasonal Monsoon winds or Trade winds).

Q: Which place in India/world receives the highest rainfall, and what share of India's total rainfall comes from the Southwest Monsoon?

A: Mawsynram (Meghalaya) receives the highest rainfall in the world (~1,141 cm annually). The Southwest Monsoon accounts for roughly 75% of India's total rainfall.

MOCK TEST (WITH ANSWERS)

1. A sudden fall in barometer reading most likely indicates approaching:

A. Clear weather

B. Rain

C. A storm ✓

D. An earthquake

Explanation: A storm. [easy, web-research]

2. Which instrument is used to measure humidity?

A. Hydrometer

B. Altimeter

C. Psychrometer ✓

D. Radiometer

Explanation: Psychrometer. [medium, web-research]

3. North India's continental-type climate (large seasonal temperature swings) is mainly due to:

A. Vertical sun rays

B. Its location far from the sea ✓

C. Its flat plains

D. High altitude

Explanation: Distance from the moderating influence of the seas. [medium, web-research]

4. Which Indian states are most affected by tropical cyclones?

A. Tamil Nadu and Kerala

B. Odisha and Andhra Pradesh ✓

C. Karnataka and Odisha

D. Kerala and Karnataka

Explanation: Odisha and Andhra Pradesh. [easy, web-research]

5. Cyclones in the Bay of Bengal and Arabian Sea form only in the Bay of Bengal — true or false?

A. True — Arabian Sea cyclones don't occur

B. False — cyclones form in both, just more often in the Bay of Bengal ✓

C. True, but only in winter

D. False — they only form in the Arabian Sea

Explanation: Cyclones form in both seas; the Bay of Bengal simply sees them more frequently/intensely. [medium, question-bank-2026]

6. The Chinook wind is associated with which region?

A. Africa

B. Thailand

C. North America ✓

D. Russia

Explanation: North America. [medium, web-research]

7. Per the World Meteorological Organisation, a storm is classified as a cyclone once wind speed exceeds:

A. 50 kph

B. 63 kph ✓

C. 75 kph

D. 100 kph

Explanation: 63 kph. [medium, web-research]

8. 'Twisters' is another name for:

A. Tsunamis

B. Landslides

C. Earthquakes

D. Tornadoes ✓

Explanation: Tornadoes. [easy, web-research]

9. Winds that blow consistently from one direction throughout the year are called:

A. Monsoon winds

B. Trade winds

C. Planetary winds ✓

D. Polar winds

Explanation: Planetary winds. [medium, web-research]

10. Which place receives the highest rainfall in the world?

A. Cherrapunji

B. Mawsynram ✓

C. Shillong

D. Darjeeling

Explanation: Mawsynram, Meghalaya. [medium, web-research]

Part B – Aptitude & Mental Ability: Time and Work

READING MATERIAL

Time and Work is the last of the 10 named Aptitude skills — after this, Part B moves to its Reasoning unit. Real exam evidence shows heavy use of 'trap' questions designed to look like they need complex math but actually resolve via a simple ratio.

Q: Method — inverse proportion (more workers, less time): 12 men complete a work in 36 days. In how many days will 18 men finish the same work?

A: Men × Days = constant (total work). 12×36 = 18×x → x = 432/18 = 24 days. More workers means proportionally fewer days.

Q: Method — sequential combined work: A, B, C can complete a job alone in 8, 12, 16 days respectively. A and C work together for 2 days, then C leaves and B joins A. In how many more days do A and B finish?

A: Work rates: A=1/8, B=1/12, C=1/16 per day. In 2 days, A+C complete 2×(1/8+1/16)=2×(3/16)=3/8 of the work. Remaining = 5/8. A+B's combined rate = 1/8+1/12 = 5/24 per day. Days needed = (5/8)÷(5/24) = (5/8)×(24/5) = 3 days.

Q: Method — wages split by share of work done: A can do a job in 10 days, B in 15 days. They work together for 5 days, then C finishes the remaining work in 2 days. Total payment for the whole job is Rs.5,000 — how should it be split?

A: Pay is split in proportion to how much of the TOTAL work each person actually did, not how many days they worked. In 5 days, A completes 5/10=1/2 of the job, B completes 5/15=1/3; together that's 5/6, leaving 1/6 for C to finish. Split of Rs.5,000 by work share: A gets 1/2×5000=Rs.2,500; B gets 1/3×5000≈Rs.1,667; C gets 1/6×5000≈Rs.833. The key insight is the split follows WORK DONE, not days worked — C worked fewer days than B but did a comparable or different share depending on the rates, and pay tracks output, not time.

Q: Method — pairwise combined rates → solve for all three together: A+B finish a job in 12 days, B+C in 15 days, C+A in 20 days. In how many days can A, B, C finish it together?

A: Add all three pairwise rates: (A+B)+(B+C)+(C+A) = 1/12+1/15+1/20 = 2(A+B+C)'s rate. LCM=60: 5/60+4/60+3/60=12/60=1/5. So 2×(A+B+C rate)=1/5 → (A+B+C) rate=1/10 → together they finish in 10 days.

Q: Trap question: If 9 spiders make 9 webs in 9 days, how many days does 1 spider take to make 1 web?

A: 9 days — NOT 1 day. Each spider makes 1 web in 9 days regardless of how many spiders work in parallel (they're not collaborating on the SAME web). This tests whether you recognize the rate-per-individual stays fixed, the classic trap being to assume the answer scales down to 1.

Q: Method — pipes filling and draining together: Two pipes fill a tank in 30 and 40 minutes separately; a third pipe drains it in 24 minutes. With all three open, how long to fill the tank?

A: Combined rate = 1/30+1/40−1/24 (subtract the draining pipe's rate). LCM=120: 4/120+3/120−5/120 = 2/120 = 1/60. So the tank fills in 60 minutes.

Q: Method — provisions lasting fewer days with more people (inverse proportion again): 1,000 soldiers have 70 days of provisions. If 400 more soldiers join, how long will provisions last?

A: People × Days = constant (total provisions). 1000×70 = 1400×x → x = 70000/1400 = 50 days.

Q: Method — find the second worker's solo time from a combined time and the first worker's solo time: A and B together finish a job in 16 days; A alone takes 48 days. How long does B alone take?

A: B's rate = (A+B)'s rate − A's rate = 1/16 − 1/48 = 3/48−1/48 = 2/48 = 1/24. So B alone takes 24 days.

Q: Trap question: If 5 persons complete 5 projects in 5 days, how long do 50 persons take to complete 50 projects?

A: Still 5 days — NOT 50 days. The ratio of people to projects stays the same (1:1), so the per-project time doesn't change; 50 people working on 50 projects is just 10 independent copies of the original 5-person/5-project scenario happening in parallel.

MOCK TEST (WITH ANSWERS)

1. 15 men complete a work in 24 days. In how many days will 20 men finish the same work?

A. 16 days

B. 18 days ✓

C. 20 days

D. 30 days

Explanation: 15×24=20×x → x=360/20=18 days. [easy, authored]

2. A and B together can finish a job in 18 days; A alone takes 45 days. How long does B alone take?

A. 24 days

B. 27 days

C. 30 days ✓

D. 36 days

Explanation: B's rate = 1/18−1/45 = 5/90−2/90=3/90=1/30. B alone takes 30 days. [medium, authored]

3. If 7 workers build 7 walls in 7 days, how many days does 1 worker take to build 1 wall?

A. 1 day

B. 7 days ✓

C. 14 days

D. 49 days

Explanation: 7 days — each worker's individual rate is unaffected by how many work in parallel on separate walls. [medium, authored]

4. Two pipes fill a tank in 12 and 15 minutes separately; a third pipe drains it in 20 minutes. With all three open, how long to fill the tank?

A. 8 minutes

B. 10 minutes ✓

C. 12 minutes

D. 15 minutes

Explanation: Rate = 1/12+1/15−1/20. LCM=60: 5/60+4/60−3/60=6/60=1/10. Time = 10 minutes. [medium, authored]

5. 800 soldiers have provisions for 60 days. If 200 more soldiers join, how long will the provisions last?

A. 40 days

B. 45 days

C. 48 days ✓

D. 50 days

Explanation: 800×60=1000×x → x=48000/1000=48 days. [easy, authored]

6. A, B, and C can do a job alone in 24, 30, and 40 days respectively. Working together, how long will they take?

A. 8 days

B. 10 days ✓

C. 12 days

D. 15 days

Explanation: Combined rate = 1/24+1/30+1/40. LCM=120: 5/120+4/120+3/120=12/120=1/10. Together they take 10 days. [medium, authored]

7. If 8 people complete 8 tasks in 8 days, how long do 40 people take to complete 40 tasks?

A. 8 days ✓

B. 20 days

C. 40 days

D. 320 days

Explanation: Still 8 days — the people-to-task ratio (1:1) is unchanged, so per-task time is unchanged. [medium, authored]

Part C – General English: Poem: "Special Hero" — Christine M. Kerschan

READING MATERIAL

A poem dedicated to the poet's father, calling him her 'special hero' for his love and protection. Real coaching-site evidence tests figure-of-speech identification per line — and uniquely among the poems covered so far, several DIFFERENT devices appear across just 6 short lines, so precision matters more than pattern-guessing here.

Q: What is "Special Hero" about?

A: The poet's tribute to her father — recalling how he held her as a baby, gave her love and protection, and reflecting on how lucky she feels that he was 'chosen' to be her father.

Q: Identify the figure of speech: "There is something special about a father's love."

A: Metaphor.

Q: Identify the figure of speech: "Seems it was sent to me from someplace up above."

A: Hyperbole — an exaggerated, larger-than-literal claim (the love being literally 'sent from above') used for emotional emphasis, not meant literally.

Q: Identify the figure of speech: "How did I get so lucky, you were the dad chosen for me?"

A: Rhetorical question — asked for emotional effect, not expecting a literal answer.

Q: What figure of speech is the repetition of "I just wanted you to know"?

A: Anaphora — repetition of the same phrase at the start of successive lines.

Q: Identify the figure of speech: "Our love is everlasting."

A: Hyperbole — an absolute, exaggerated claim ('everlasting') used to emphasize depth of feeling.

MOCK TEST (WITH ANSWERS)

1. "Special Hero" is dedicated to whom, and written by whom?

A. Dedicated to a friend, by Sarojini Naidu

B. Dedicated to her father, by Christine M. Kerschan ✓

C. Dedicated to a teacher, by David Bates

D. Dedicated to a soldier, by Edgar Albert Guest

Explanation: Dedicated to the poet's father, written by Christine M. Kerschan. [easy, web-research]

2. Identify the figure of speech: "There is something special about a father's love."

A. Simile

B. Metaphor ✓

C. Personification

D. Hyperbole

Explanation: Metaphor. [medium, web-research]

3. Identify the figure of speech: "Seems it was sent to me from someplace up above."

A. Anaphora

B. Alliteration

C. Hyperbole ✓

D. Personification

Explanation: Hyperbole — an exaggerated, emotionally emphatic claim. [medium, web-research]

4. "How did I get so lucky, you were the dad chosen for me?" is an example of:

A. Metaphor

B. Simile

C. Personification

D. Rhetorical question ✓

Explanation: A rhetorical question, asked for effect rather than a literal answer. [medium, web-research]

5. The repeated phrase "I just wanted you to know" is an example of:

A. Anaphora ✓

B. Epizeuxis

C. Anadiplosis

D. Chiasmus

Explanation: Anaphora — repetition at the start of successive lines. [hard, web-research]

6. "Our love is everlasting" is an example of:

A. Hyperbole ✓

B. Metaphor

C. Simile

D. Alliteration

Explanation: Hyperbole — an absolute, exaggerated claim. [medium, web-research]